

THOMAS SIRCHI

Italy | +39 334 999 4980 | thomasirchi@gmail.com

PROFILE

Computational biologist with experience in single-cell and multi-omics analysis of cancer biology. Skilled in scRNA-seq, bulk RNA-seq, miRNA-seq, DNA methylation, immune deconvolution, machine learning, and statistical modelling. Experienced in investigating tumour cell plasticity, malignant state transitions, immune phenotypes, genomic instability, and treatment response across lung cancer and haematological malignancies. Strong background in building reproducible R and Python workflows and translating high-dimensional molecular data into biologically and interpretable findings.

EXPERIENCE

Bioinformatician – Functional Oncology | Casa Sollievo della Sofferenza IRCCS

November 2025 – present

- scRNA-seq analysis of tumour plasticity and state transitions in lung adenocarcinoma using trajectory inference and RNA velocity.
- Multi-omics integration of miRNA-seq, bulk RNA-seq, methylation, immune deconvolution, and DNA alterations to link regulatory signals to tumour phenotypes (MOFA).
- ML and statistical modelling of immune activation, PD-L1/IFNG phenotypes, and clinical variables for biomarker discovery.

Master's Thesis – Computational oncology | University of Trento

September 2024 – February 2025

- Built a reference atlas of neutrophil subtypes by integrating methylation and RNA-seq data.
- Immune deconvolution across multi-cancer cohorts; TANs signatures associating with prognosis and genomic instability.
- Immune-derived cfDNA signals as candidates for non-invasive biomarker development.

ML Research – Paediatric ALL Subtype Classification | BITS 2024 | 36th Pezcoller Symposium

June 2024 – 2025

- XGBoost and neural-network models for subtype classification from large-scale genomic data; SHAP-based feature interpretation.
- Presented findings at two national bioinformatics and cancer research meetings.

Biotechnologist – Genotyping | MRC Harwell, Mary Lyon Centre

June – September 2022

- High-throughput TaqMan RT-PCR genotyping for engineered mouse model workflow.

EDUCATION

MSc Quantitative and Computational Biology, University of Trento, July 2025

BSc Biological Sciences, University of Perugia, February 2022

TECHNICAL SKILLS

- **Programming:** Python | R | Bash
- **Frameworks:** Seurat | Scanpy | tidyverse | DESeq2 | scikit-learn | PyTorch
- **Domains:** scRNA-seq | bulk RNA/miRNA-seq | multi-omics integration | ML | statistical modelling | HPC workflows

LANGUAGES

- Italian (native)
- English (C1+)